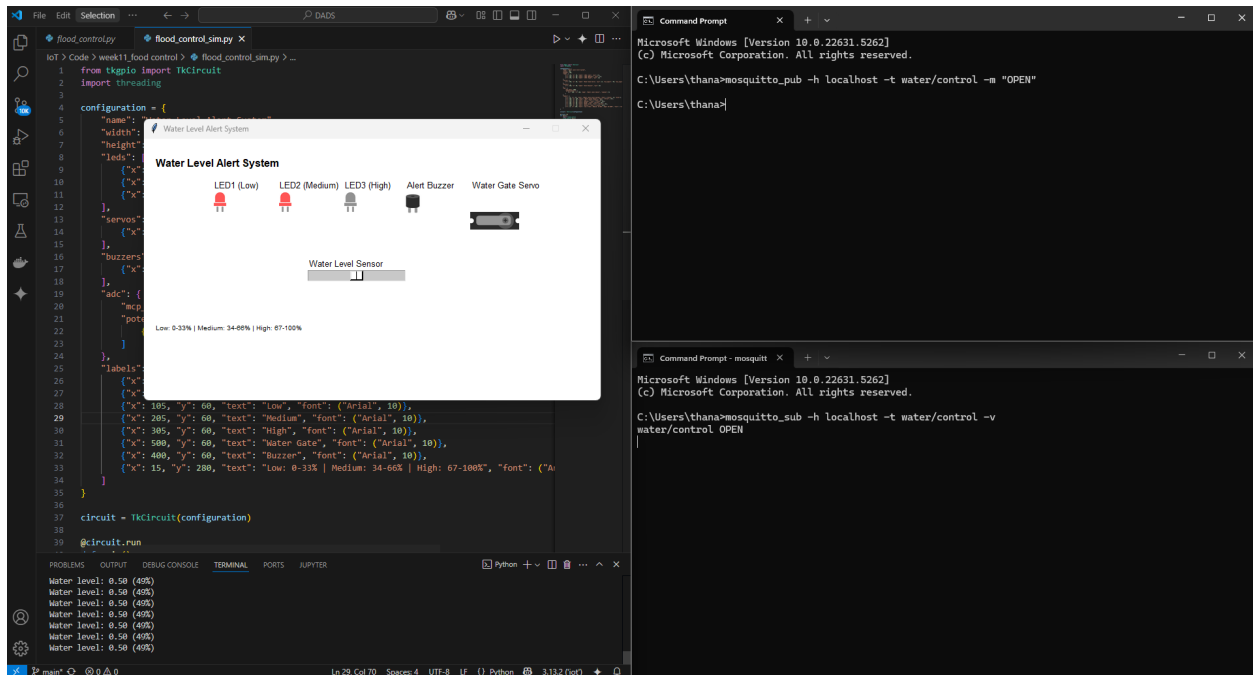
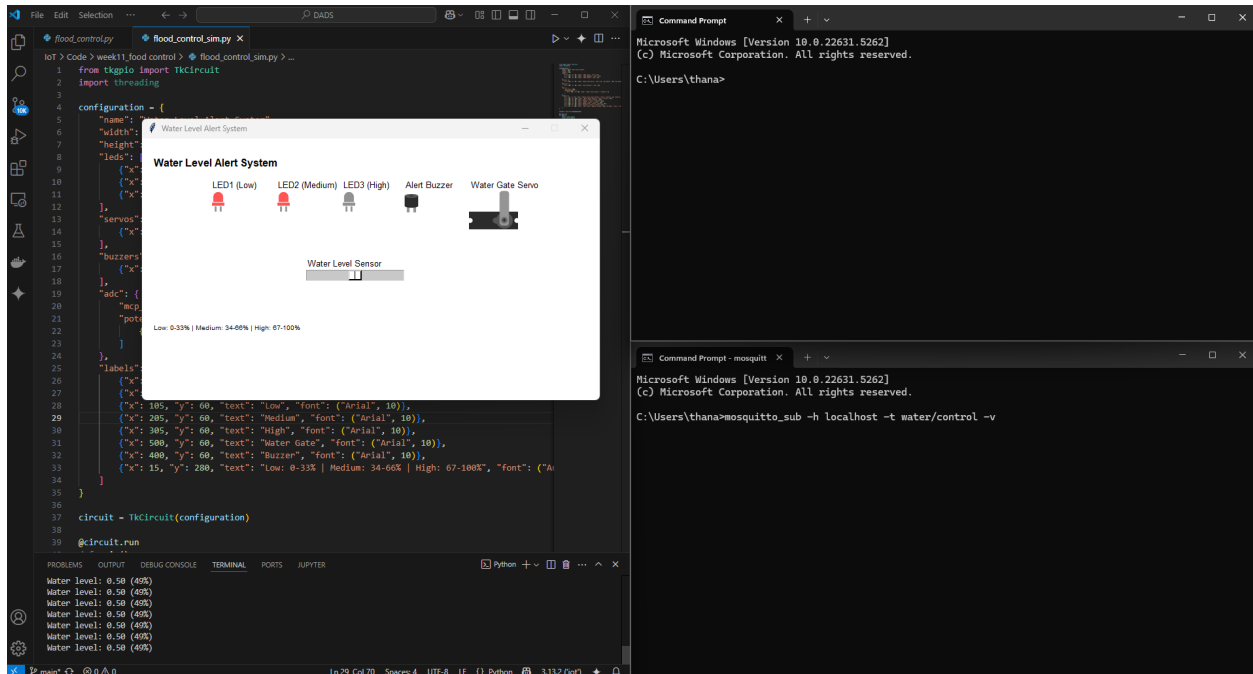
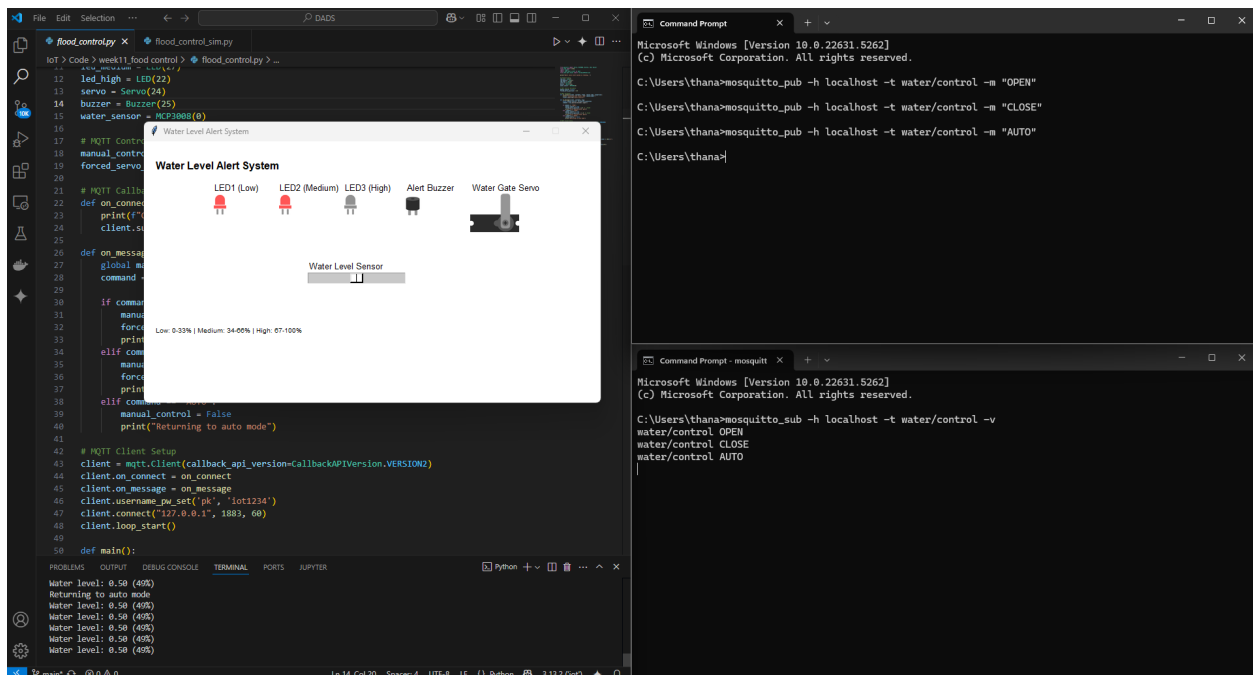
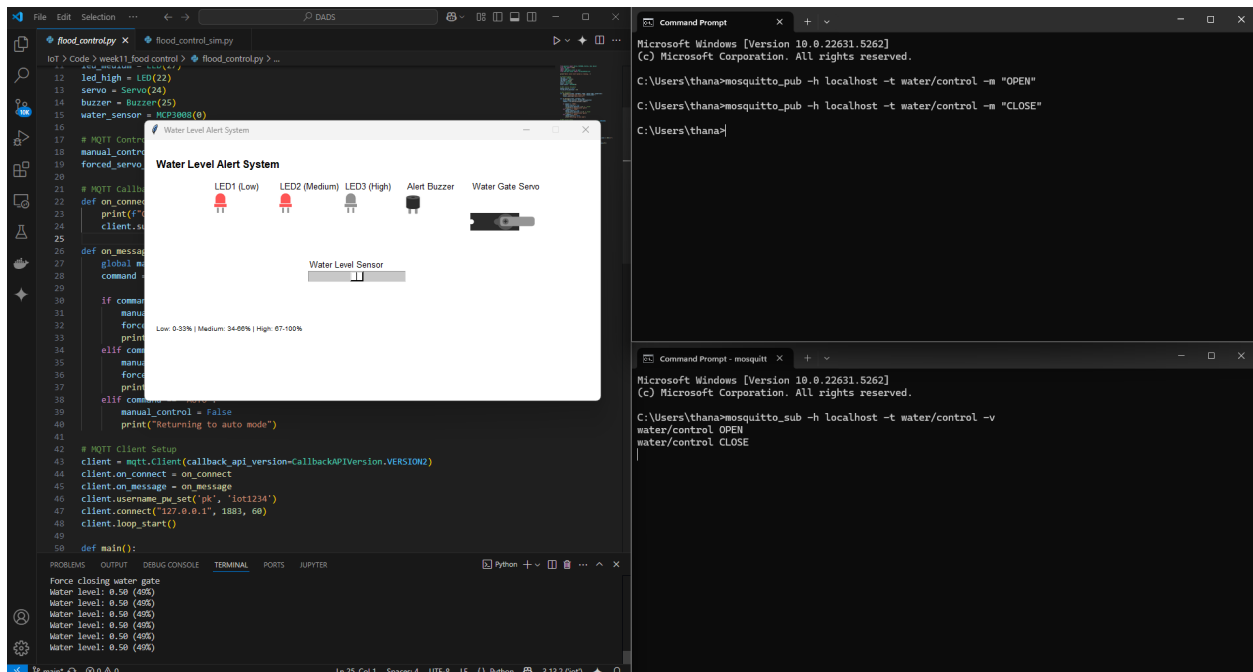






flood_control.py

```

from gpiozero import Servo, MCP3008, Button, LED, Buzzer
from time import sleep
import signal
import paho.mqtt.client as mqtt
from paho.mqtt.enums import CallbackAPIVersion

```

```

print("Water Level Alert System is running...")

# Hardware Setup
led_low = LED(17)
led_medium = LED(27)
led_high = LED(22)
servo = Servo(24)
buzzer = Buzzer(25)
water_sensor = MCP3008(0)

# MQTT Control Variables
manual_control = False
forced_servo_position = 0.0

# MQTT Callbacks
def on_connect(client, userdata, flags, reason_code, properties):
    print(f'Connected with result code {reason_code}')
    client.subscribe("water/control")

def on_message(client, userdata, msg):
    global manual_control, forced_servo_position
    command = msg.payload.decode().upper()

    if command == "OPEN":
        manual_control = True
        forced_servo_position = 1.0 # เปิดสุด
        print("Force opening water gate")
    elif command == "CLOSE":
        manual_control = True
        forced_servo_position = -1.0 # ปิดสุด
        print("Force closing water gate")
    elif command == "AUTO":
        manual_control = False
        print("Returning to auto mode")

# MQTT Client Setup
client = mqtt.Client(callback_api_version=CallbackAPIVersion.VERSION2)
client.on_connect = on_connect
client.on_message = on_message
client.username_pw_set('pk', 'iot1234')
client.connect("127.0.0.1", 1883, 60)
client.loop_start()

def main():
    print("System active. Press Ctrl+C to exit.")
    try:
        while True:
            water_level = water_sensor.value

            print(f'Water level: {water_level:.2f} ( {int(water_level * 100)}% )')

```

```

# ภูมิพล
led_low.on() if water_level >= 0 else led_low.off()
led_medium.on() if water_level > 0.33 else led_medium.off()

if water_level > 0.66:
    led_high.on()
    buzzer.on()
else:
    led_high.off()
    buzzer.off()

# ภูมิ Servo
if manual_control:
    servo.value = forced_servo_position
else:
    servo_position = (2 * water_level) - 1
    servo.value = servo_position

sleep(0.5)

except KeyboardInterrupt:
    client.loop_stop()
    print("\nSystem stopped")

if __name__ == "__main__":
    main()

```

flood_control_sim.py

```

from tkgpio import TkCircuit
import threading

configuration = {
    "name": "Water Level Alert System",
    "width": 700,
    "height": 400,
    "leds": [
        {"x": 105, "y": 80, "name": "LED1 (Low)", "pin": 17},
        {"x": 205, "y": 80, "name": "LED2 (Medium)", "pin": 27},
        {"x": 305, "y": 80, "name": "LED3 (High)", "pin": 22}
    ],
    "servos": [
        {"x": 500, "y": 80, "name": "Water Gate Servo", "pin": 24, "min_angle": -90, "max_angle": 90,
"initial_angle": 0}
    ],
    "buzzers": [
        {"x": 400, "y": 80, "name": "Alert Buzzer", "pin": 25}
    ],
    "adc": {

```

```

    "mcp_chip": 3008,
    "potentiometers": [
        {"x": 250, "y": 200, "name": "Water Level Sensor", "channel": 0}
    ]
},
"labels": [
    {"x": 15, "y": 25, "text": "Water Level Alert System", "font": ("Arial", 16, "bold")},
    {"x": 250, "y": 180, "text": "Water Level Sensor", "font": ("Arial", 10)},
    {"x": 105, "y": 60, "text": "Low", "font": ("Arial", 10)},
    {"x": 205, "y": 60, "text": "Medium", "font": ("Arial", 10)},
    {"x": 305, "y": 60, "text": "High", "font": ("Arial", 10)},
    {"x": 500, "y": 60, "text": "Water Gate", "font": ("Arial", 10)},
    {"x": 400, "y": 60, "text": "Buzzer", "font": ("Arial", 10)},
    {"x": 15, "y": 280, "text": "Low: 0-33% | Medium: 34-66% | High: 67-100%", "font": ("Arial",
10)}
]
}

circuit = TkCircuit(configuration)

@circuit.run
def main():
    import flood_control
    flood_control.main()

```